Comparing data sets

1 The number of goals scored by Team 1 and Team 2 in a football tournament are recorded in the following table:

Match	Team 1	Team 2
A	2	3
B	4	2
C	5	2
D	3	5
E	3	4

a Find the total number of goals scored by both teams in Match C.

b Find the total number of goals scored by Team 1 across all the matches.

c Calculate the mean number of goals scored by:

i Team 1 **ii** Team 2

2 The table shows the scores of Student A and Student B in 5 separate tests:

Test	Student A	Student B
1	79	77
2	74	83
3	86	72
4	89	94
5	98	97

a Find the mean of the scores of:

i Student A **ii** Student B

b What is the mean of the combined scores of the two students?

c What is the highest score overall and which student obtained that score?

d What is the lowest score overall and which student obtained that score?

3 The salaries of men and women working the same job at the same company are given below:

Men	80 000	80 000	75 000	80 000	75 000	70 000	80 000
Women	70 000	70 000	75 000	70 000	70 000	80 000	75 000

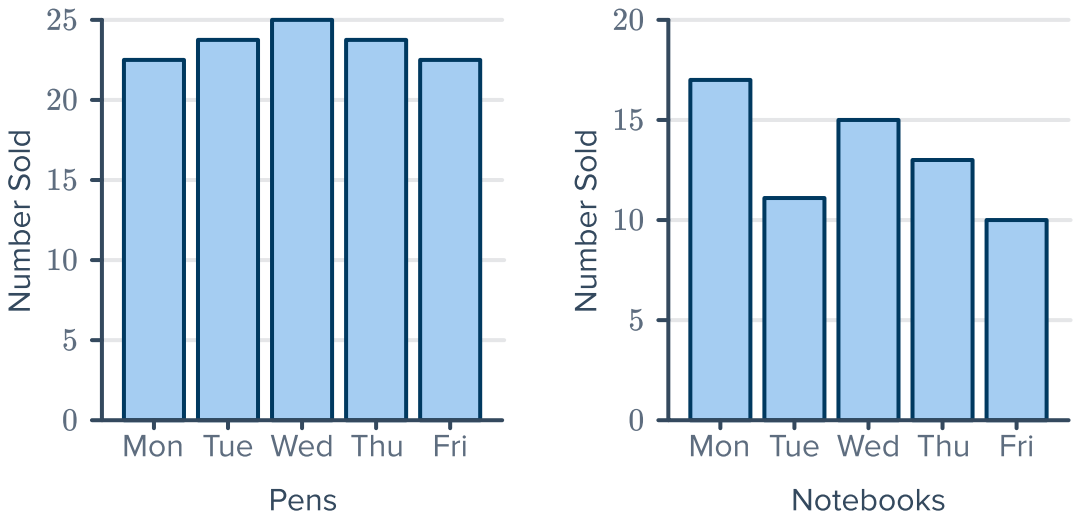
a Copy and complete the table, giving your answers correct to two decimal place where appropriate:

	Men	Women
Mean		
Median		
Mode		
Range		

b Which gender seems to be paid better? Justify your answer.

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4 A stationery shop kept a weekly record of the number of pens and notebooks sold:



a Find the number of pens sold on Tuesday.

b Find the number of notebooks sold on Friday.

c Find the total number of notebooks sold during the week.

d Find the total number of pens sold during the week

e Find the percentage of pens sold on Thursday, rouding your answer to two decimal places.

f Calculate the mean number of notebooks sold per day. Round your answer to one decimal place.

g Calculate the mean number of pens sold per day. Round your answer to one decimal place.

h Which is the better selling product? Justify your answer.

5 The runs scored by each player in a cricket match are displayed below:

- Team A: $63, 0, 81, 22, 39, 66, 9, 18, 3, 3, 5$
- Team B: $21, 26, 18, 69, 62, 26, 5, 28, 2, 1, 2$

a Find the mean runs of the following teams, rounded to one decimal place:

i Team A

ii Team B

b Which team scored more runs?

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6 In a standardised test out of 100, the students in two classes got the results displayed below:

- Class A: $70, 64, 94, 78, 79, 61, 69, 93, 64, 79, 64, 71, 82, 68, 66$
- Class B: $86, 68, 51, 80, 63, 87, 79, 89, 62, 79, 84, 77, 67, 80, 75$

a Find the median mark for:

i Class A **ii** Class B

b Using the medians, which class did better on the test?

7 The monthly rainfall for two cities across a year is displayed below:

- City A: $122, 120, 121, 67, 79, 24, 8, 46, 27, 67, 80, 128$
- City B: $51, 66, 33, 78, 79, 92, 78, 99, 41, 60, 26, 56$

a Find the range in monthly rainfall for:

i City A **ii** City B

b Which city had more consistent rainfall throughout the year?

8 Marge grows two different types of bean plants. She records the number of beans that she picks from each plant for 10 days. Her records are shown below:

- Plant A: $10, 4, 4, 5, 7, 10, 3, 3, 9, 10$
- Plant B: $8, 7, 5, 5, 9, 7, 8, 7, 5, 6$

a What is the mean number of beans picked per day for Plant A? Round your answer to one decimal place.

b What is the mean number of beans picked per day for Plant B? Round your answer to one decimal place.

c What is the range for Plant A?

d What is the range for Plant B?

e Which plant produces more beans on average?

f Which plant has a more consistent yield of beans?

9 The weight (in kilograms) of a group of men and women were recorded and presented in a stem and leaf plot as shown:

Women		Men
7 6 3	5	
8 6 3 1 1	6	2 3 3 8 9
3 1	7	1 2 4 8
	8	3

a Find the mean weight of the group of men.

b Find the mean weight of the group of women.

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c Which group is heavier?

Key: $8|3 = 83$

10 The stem and leaf plot shows the batting scores of two cricket teams, India and Sri Lanka:

a Find the highest score in:

i India **ii** Sri Lanka

b Find the mean score of:

i India **ii** Sri Lanka

c Calculate the combined mean of the two teams.

India		Sri Lanka
9 6	3	0 2 5 6 9
9 7 3 1 1	4	2 7
7 2 0	5	6 8
	6	3

Key: $8|3 = 83$

11 The stem and leaf plot shows the test scores of two classes, A and B:

a Find the highest score in:

i Class A **ii** Class B

b Find the mean score of:

i Class A **ii** Class B

c Calculate the combined mean of the two classes, correct to two decimal places.

Class A		Class B
7 3 2	6	2 7
9 8 7 1 0	7	1 3 6 7
3 0	8	1 2 4 6
	9	1 8

Key: $8|3 = 83$

12 The stem and leaf plot shows the number of books read in a year by a random sample of university and high school students:

a Interpret the lowest score for the University students.

b Compare the medians of both groups of students.

c For which student group(s) is the mean greater than the median?

Univerity Students		High School Students
7	0	
6 6 3	1	0 0 3 5
4 3 2 1	2	1 2 4 4 6
9 8 8 6	3	1 8 9
8 2	4	0 1
	5	
	6	
3	7	

Key: $4|1|2 = 14$ books and 12 books

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13 The back to back stem and leaf plot shows the amount of cash (in dollars) carried by a random sample of teenage boys and girls:

a Which group carried more cash?

b Find the median amount of cash that the boys carried

c Find the median amount of cash that the girls carried.

d Which group's distribution is roughly bell shaped?

e Which group has more variation in the amounts of cash?

f Were there any outliers in the boys' amounts? If so, what are the value(s)?

g Were there any outliers in the girls' amounts? If so, what are the value(s)?

Boys		Girls
7	0	
1	1	1
5 4 1	2	2 6 8
8 5 4	3	3 4 4 6 6 8 9
9 8 2 2 2 1	4	3 4 6
9 7 4 3	5	4
8 5 2	6	
3 1	7	

Key: $1|2|2 = \$21$ and $\$22$

14 The following back to back stem and leaf plot shows the length (in minutes) of a random sample of phone calls made by Sharon and Tricia:

a Who made a 14 minute phone call?

b Who has the higher median?

c Is Sharon's mean greater than her median?

d Is Tricia's mean greater than her median?

Sharon		Tricia
3	1	3 4
7 6 4 3 2	2	6 7 8
9 8	3	2 4
4 3	4	1 2
7 6	5	6 7 8

Key: $2|2|6 = 22$ and 26

15 The back to back stem and leaf plot shows the number of pieces of paper used over several days by Charlie's and Dylan's students:

a Did Charlie's students use 7 pieces of paper on any day?

b Who's class had the higher median?

c Is the median greater than the mean in both groups?

Charlie's students		Dylan's students
7	0	7
3 2 1	1	3
8	2	8
4 3 2	3	3 4
9	4	5 6 7
2	5	2 3

Key: $1|1|3 = 11$ and 13

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16 The back to back stem and leaf plot shows the number of desserts ordered at Hotel A and Hotel B over several randomly chosen days:

a Interpret the lowest score for Hotel A.

b Which hotel's median is higher?

c Is the mean greater than the median in both groups?

Hotel A		Hotel B
3	0	
4 3 2	1	3 4
7 6	2	7
4 3	3	3 4
6	4	6 7
2	5	2 3 4

Key: $2|1|3 = 12 \text{ and } 13$

17 The median house price in Oklachusetts is $\$750\,000$ with a mean price of $\$800\,000$ and the median house price in Brockway is $\$750\,000$ with a mean price of $\$680\,000$.

Which town is most likely to have some very expensive houses? Explain your answer.

18 Two Chemistry classes, each with 15 students, sit a 10-question multiple choice test, each with four possible answers (only one of which is correct). Their class results, out of 10, are below:

- Class 1: $3, 2, 1, 1, 2, 5, 2, 2, 2, 3, 2, 1, 5, 5, 4$
- Class 2: $10, 6, 9, 8, 10, 9, 7, 7, 6, 8, 8, 10, 8, 9, 8$

a Complete the following table, rounding your answers to one decimal place when necessary:

	Mean	Median	Mode	Range
Class 1				
Class 2				

b Which class was more likely to have studied for their test? Explain your answer.

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19 The hours of sleep per night for two people over a two week period are shown below:

- Person A: $10, 9, 6, 6, 7, 8, 5, 6, 9, 9, 7, 5, 10, 9$
- Person B: $7.5, 7.5, 8, 7.5, 7, 8, 7.5, 7, 7, 7, 7, 7, 8, 8$

a Complete the following table, rounding your answers to one decimal place when necessary:

	Mean	Median	Mode	Range
Person A				
Person B				

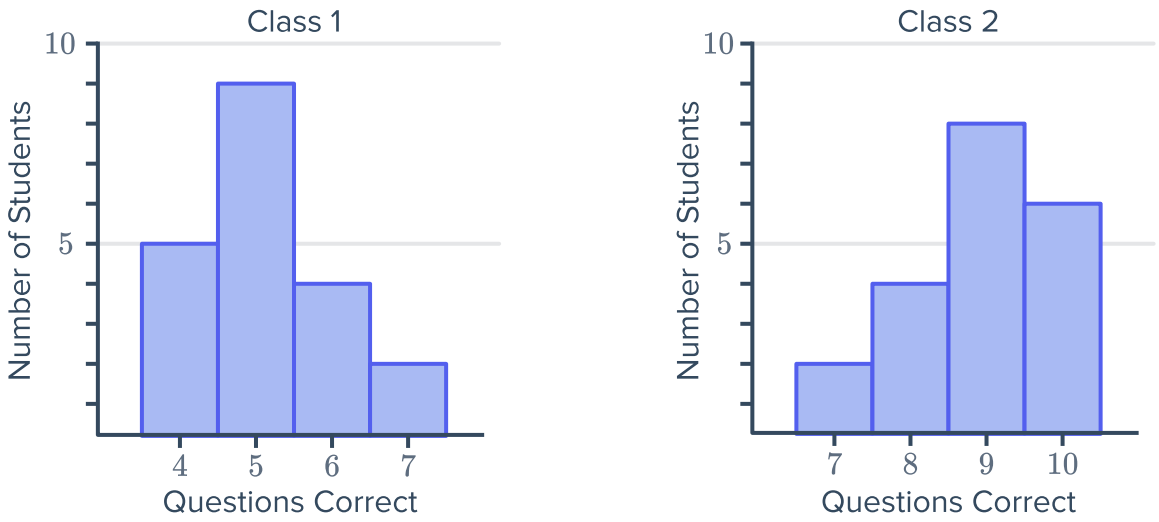
b Which person seems to be the least consistent in their sleep habits?

c Which statistical piece of evidence supports your answer in part (b)?

d Which person had the most sleep over the 14 nights?

e Which statistical piece of evidence supports your answer in part (d)?

20 Two Science classes, each with 20 students, were given a 10 question True/False test. The results for each class are shown below:



a Calculate the mean for Class 1 correct to one decimal place.

b Calculate the mean for Class 2 correct to one decimal place.

c Find the range for Class 1.

d Find the range for Class 2.

e Do you think Class 1 studied for their test? Justify your answer.

f Do you think Class 2 studied for their test? Justify your answer.

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21 The ages of employees at two competing fast food restaurants on a Saturday night are recorded. Some statistics are given in the following table:

	Mean	Median	Range
Berger's Burgers	18	17	8
Fry's Fries	18	19	4

a If the data for each restaurant was represented using a histogram, what would the likely shape of the histogram for Berger's Burgers be?

b Which restaurant likely has the oldest employee on the night the data is recorded?

c Which restaurant has the most employees of a similar age?

d Which statistical piece of evidence supports your answer to part (c)?

e Which restaurant has an older workforce?

f Which statistical piece of evidence supports your answer to part (e)?

22 The beaks of two groups of bird are measured, in millimetres, to determine whether they might be of the same species. The results are presented below:

- Group 1: $46, 34, 35, 44, 31, 37, 38, 31, 40, 29$
- Group 2: $44, 41, 40, 51, 52, 41, 36, 51, 51, 38$

a Calculate the range for:

i Group 1 **ii** Group 2

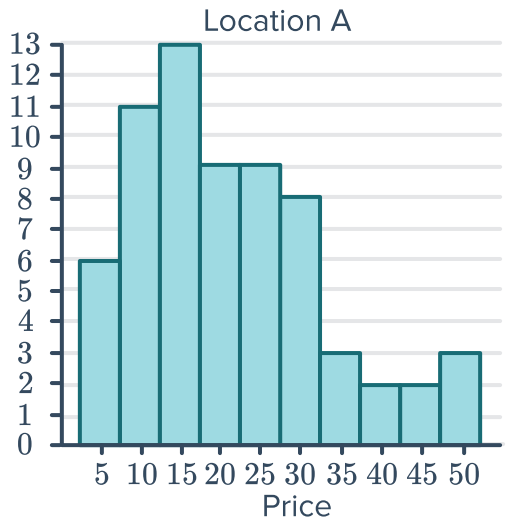
b Calculate the mean, rounded to one decimal place, for:

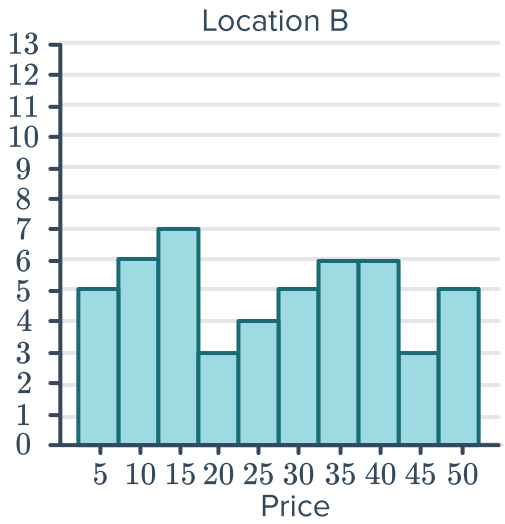
i Group 1 **ii** Group 2

c Do you think the two groups of birds are from the same species? Explain your answer.

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23 The prices of sales at two locations of a shop are recorded in the histograms below:





a Find the mean sale price for:

i Location A

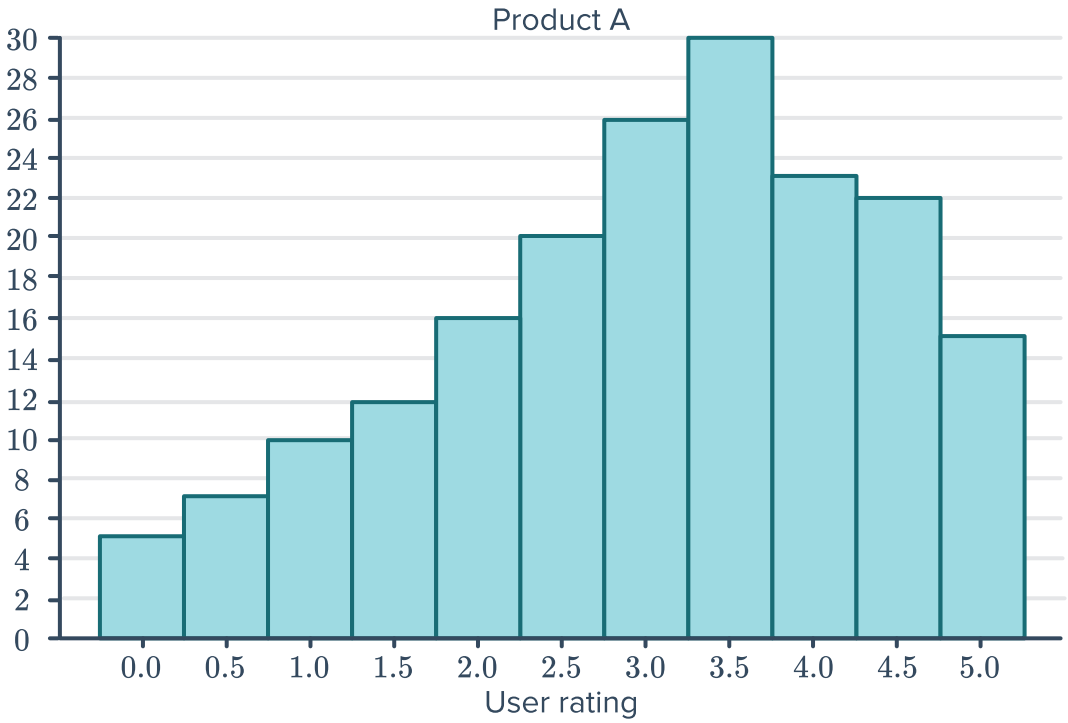
ii Location B

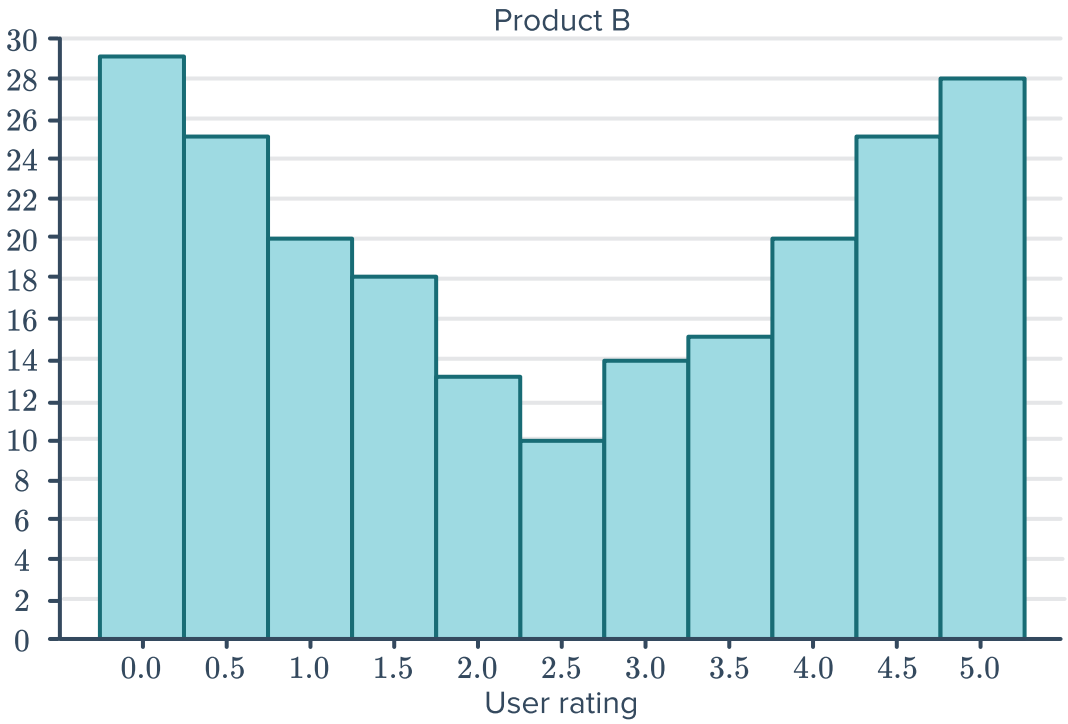
b Which location made more sales?

c Which location had a higher average sale price?

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24 An online shopping website records the user ratings for two similar products in the histograms below:





a Find the median user rating for:

i Product A

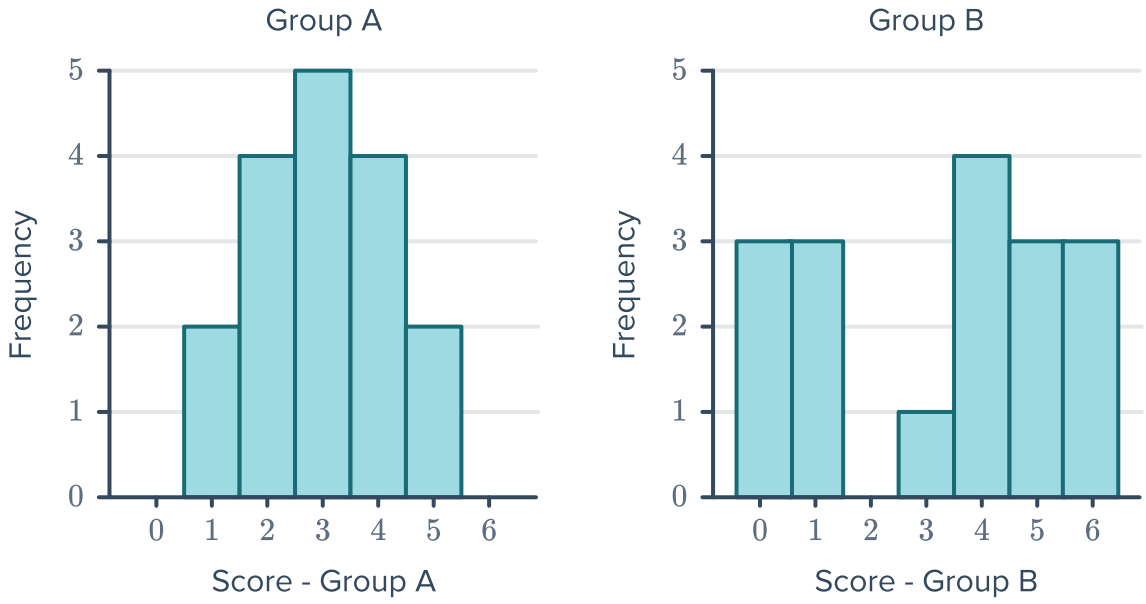
ii Product B

b According to the website's rating system, a perfect rating is 5.0. Which product got more perfect ratings?

c According to the website's rating system, a positive rating is greater than 3. Which product got more positive ratings?

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25 The following histograms show the season results of two soccer groups, Group A and Group B, and the number of games (frequency) in which they scored a certain number of goals:



a Find the mode for Group A.

b Find the mode for Group B.

c Find the range for Group A.

d Find the range for Group B.

e Which group scored the lowest total number of goals during the season?

f Which group has the most varied results?

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